

Co-Speech Motion for Virtual Agents in Dialogue Using LLM-Driven Primitive Action Selection

Muhammad Yeza Baihaqi^{1,2,3}, Angel García Contreras^{2,1}, Seiya Kawano^{4,2}, Koichiro Yoshino^{3,2,1}

¹Nara Institute of Science and Technology, Japan; ²Guardian Robot Project, RIKEN, Japan;

³Institute of Science Tokyo, Japan; ⁴Kyoto Institute of Technology, Japan

muhammad.yeza.baihaqi.lx2@naist.ac.jp, angel.garciacontreras@riken.jp,
seyia.kawano@riken.jp, koichiro@c.titech.ac.jp

Abstract

Non-verbal behaviors, such as co-speech motion, are essential for making artificial agents more lifelike and engaging. However, existing approaches to generating co-speech motions still face significant challenges. Rule-based systems, while capable of producing natural and engaging motions, struggle with generalizability. In contrast, the common data-driven methods can generate a wide variety of gestures but come with high costs and require substantial adaptation to be applied to agents. Leveraging the power of Large Language Models (LLMs) for contextual planning and understanding, we propose an LLM-based motion control model that uses a primitive action selection strategy. This approach is expected to provide a more flexible and scalable solution for generating contextually appropriate co-speech motions across various embodied systems, including virtual agents and robots.

Index Terms: Co-speech motion, human-agent interaction, LLM-based motion control, primitive action selection

1. Introduction

Co-speech motion or gesture is a type of non-verbal behavior that accompanies speech and enhances verbal communication through body movement. In human-to-human interaction, it is common for individuals to perform co-speech motions to provide appropriate social cues [1]. Similarly, incorporating non-verbal behaviors into artificial agents can make them more lively and engaging [2]. However, generating natural and contextually appropriate co-speech motions remains a challenge.

Existing approaches, such as rule-based systems, can produce natural and engaging co-speech motions but are highly constrained. They rely on extensive hand-crafted rules, which become increasingly difficult to maintain as complexity grows [2]. The data-driven approaches can generate a wide range of gestures, but motion capture technology is often costly [2, 3]. In addition, since motions are captured from humans, they require significant adaptation before being effectively applied to different robots or virtual agents [3].

The development of Large Language Models (LLMs) has opened new possibilities for generating co-speech motions in human-agent interaction (HAI). Unlike rule-based systems, where human experts manually define motion patterns, LLMs replace these hand-crafted rules with contextual judgment. Recent research have shown that LLMs demonstrate proficiency in contextual planning, offering flexibility and an extensive knowledge base across many tasks [4, 5]. These capabilities enable LLMs to understand utterances and generate motions that complement verbal cues in a more adaptive and scalable manner.

In this demonstration, we introduce an LLM-based motion control system utilizing a primitive action selection strat-

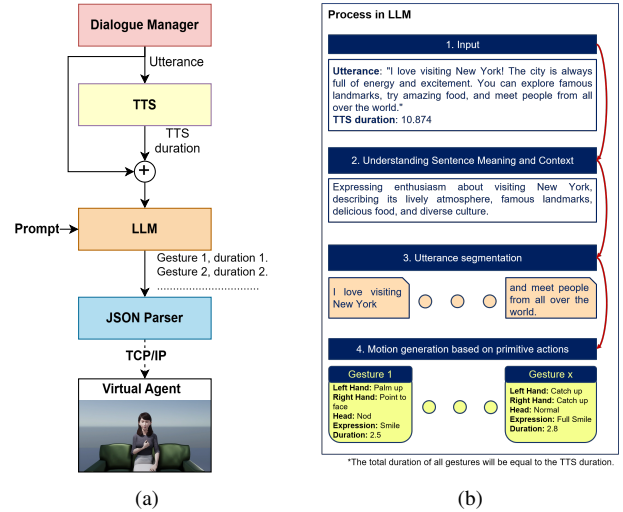


Figure 1: (a) LLM-based motion control model and (b) Motion generation in LLM

egy. Leveraging the LLM’s ability to understand the context and meaning of utterances, it selects primitive actions for each body part that form gestures, which, when performed, result in motions. Our framework guides the virtual agent ERICA in executing contextually appropriate co-speech motions.

2. Proposed methods

2.1. LLM-based motion control model

As shown in Figure 1a, the LLM-based motion control model begins by receiving an utterance input from the dialogue manager. The utterance is processed by the TTS system to obtain the speech duration. This utterance and its duration are then provided to the LLM, which generates a set of gestures, each assigned a specific duration. The total number of generated gestures (T_g) are determined by the following function:

$$T_g = \left\lceil \frac{\text{Character Count}(u)}{C} \right\rceil \quad (1)$$

where u is the utterance and C is a constant divisor. Adjusting C controls the number of gestures: smaller C generates more gestures, while larger C produces fewer. This allows the system to easily adjust the level of expressivity of the motions.

The LLM determines the distribution of duration across gestures. Some segments may be longer than others, but the total motion duration will match the TTS speech duration, ensuring smooth synchronization between speech and motion.

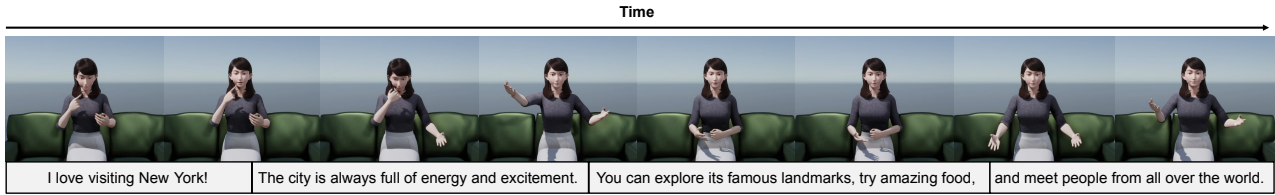


Figure 2: Example generated motions

The generated gestures are formatted in JSON and sent to the virtual agent via TCP/IP. In addition, the LLM utilized in the demonstration will be GPT-4o-2024-08-06.

2.2. LLM prompt

The prompt consists of the following components:

- **Context:** The LLM controls the virtual agent’s motions.
- **List of Actions:** The LLM selects predefined primitive actions for each body part to form a complete gesture. There are 17 actions for each arm, 5 for the head, and 3 for facial expressions. This approach is considered more robust and dynamic than setting primitive actions for the whole body. Theoretically, it can generate up to 4,335 possible gestures.
- **Instruction:** A Chain-of-Thought (CoT) approach is used to perform the task. First, the LLM must understand the meaning of the utterance and the intended message. Next, it segments the utterance based on the total number of gestures. Then, it assigns gestures from the list of predefined primitive actions for each body part to each segment, including its duration. Finally, it outputs the results in the designated format. This process is illustrated in Figure 1b.
- **Few-shot Example:** To further guide the LLM in executing the task, we provide a few-shot example in the prompt.

2.3. Multimodal virtual agent

In this demonstration, we employed a female virtual agent, ERICA [6]. For this demo, we included modules such as Amazon Polly text-to-speech (TTS)¹ for converting textual output to speech and LipSync for synchronized lip movements with speech. The gesture and facial expression modules were also included as part of the interface to receive the JSON-format output of the LLM-based motion control model into agent actions. The example generated motions are shown in Figure 2².

2.4. Preliminary performance testing

We conducted preliminary performance testing, evaluating processing time and fault analysis across 20 trials. With an average internet speed of 163.4 Mbps (download) and 115.85 Mbps (upload), the mean processing time was 1.89s (SD = 0.473). Faults, defined as the LLM selecting non-existent primitive actions, were absent in 95% of successful system runs.

3. Show and Tell Demonstration

During the demonstration, we will showcase the performance of our proposed method in live motion generation for co-speech motions. In addition to our prepared utterance examples, attendees may also provide sentence to be used for motion gen-

eration. Our approach highlights the model’s ability to generate appropriate co-speech motions dynamically for diverse inputs, without being constrained by participants’ dialogue skills or other unintended factors.

By allowing attendees to freely input their utterances—whether short or long—we can effectively demonstrate the scalability and adaptability of our method. The system dynamically selects primitive actions based on the input, ensuring that motions align naturally with the speech content. This flexibility would be harder to observe in a direct dialogue setting, as participants have two focuses: one on making a dialogue and the other on observation.

Participants will see the virtual agent animate their provided utterances with co-speech motions, offering an interactive way to evaluate the effectiveness of our approach. To accommodate participants’ preferences, we provide three language options for the agent’s speech: English, Japanese, and Dutch.

4. Conclusion

We introduced an LLM-based co-speech motion model for virtual agents using primitive action selection. This approach addresses the generalizability issue of rule-based systems and the high cost and adaptability challenges of data-driven systems. Our demonstration highlights the system’s ability to handle diverse utterances while supporting multiple languages. Preliminary testing shows that the model operates within a reasonable processing time. This work represents a promising step toward more natural and efficient co-speech motion in HAI.

5. Acknowledgements

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¹<https://aws.amazon.com/polly>

²The motion video is available at youtu.be/_luMJEPQV8